Learning stack: EFE, Dirichlet updates, and BMR I

Active-inference agents do more than infer states under a fixed model: they learn the parameters of the model, score policies by expected free energy, and revise model structure. The active-inference community has standardized all three operations (Da Costa et al. 2020; Smith et al. 2022), and Friston et al. (2024) place them at the heart of the federated belief-sharing scenario. We reimplement each in closed form so the language-acquisition, expected-free-energy, and emergence studies of sec. 5.22 rest on machine-checkable quantities rather than fitted curves.

Conjugate Dirichlet learning from co-occurrence counts I

A sentinel learns its observation model A by placing a Dirichlet prior with concentration a on each column and updating it conjugately from observation-state co-occurrence counts (Friston et al. (2024), their equations 9–12). One learning step adds the expected sufficient statistics for that step and reads off the column-normalized posterior mean,

$$a \leftarrow a + \text{counts}, \qquad \mathbb{E}[A]_{os} = \frac{a_{os}}{\sum_{o'} a_{o's}}, \tag{1}$$

implemented in `dirichlet_learning.learn_likelihood`. Intuitively, each concentration vector is a running tally of how often each outcome was seen while the creature occupied a given state: the prior seeds that tally with pseudo-counts, every step adds the co-occurrences it witnessed, and the posterior mean is simply the tally renormalized into a categorical. Likelihood learning is therefore bookkeeping — accumulate counts, then normalize — with no iterative optimization loop. We drive eq. 1 with the expected sufficient statistics under the true model — a fixed count batch count_scale $\cdot A^\star$ per step, optionally jittered by a seeded generator. As the concentrations accumulate, the expected likelihood $\mathbb{E}[A]$ converges to the data-generating $A^\star$; convergence is measured by the per-column KL divergence summed over hidden states,

Conjugate Dirichlet learning from co-occurrence counts I

$$\mathrm{KL}(A^\star \,\|\, \mathbb{E}[A]) \;=\; \sum_s \sum_o A^\star_{os} \, \ln \frac{A^\star_{os}}{\mathbb{E}[A]_{os}} \,, \tag{2}$$

which decreases monotonically toward zero — the standard-Bayes / KL fixed point. The learned likelihood always has full support (the Dirichlet prior is strictly positive), so eq. 2 is finite throughout. ISC-17 pins the monotone-decreasing KL trajectory, and the language-acquisition study of sec. 5.22 reports the descent of eq. 2 across 24 steps.

Conjugate Dirichlet learning from co-occurrence counts I

The implementation also carries the η forgetting hyperprior of Friston et al. (2024), their equation 12: before each conjugate addition the running concentration is decayed so the *total* concentration mass saturates at η rather than growing without bound, modeling an agent that stays adaptable instead of becoming infinitely confident. With η unset the classical unbounded accumulation of eq. 1 is recovered.

A sentinel scores a candidate policy π by its expected free energy $G(\pi)$, which the active-inference formulation decomposes two equivalent ways (Friston et al. (2024), their equation 2): a cost view of risk plus ambiguity, and a value view of pragmatic plus epistemic value. The two views are the same scalar rearranged, stated as eq. 18 and pinned to a zero residual by the algebraic identity eq. 19 in sec. 6. We compute every term in closed form from the categorical model (A, B, C, D_0) in `expected_free_energy.decompose`:

Expected free energy as the action-selection objective I

- *Risk* is $\mathrm{KL}(q(o \mid \pi) \,\|\, p_C(o))$, the deviation of the policy-predicted outcomes from the preferred-outcome pmf $p_C(o) = \mathrm{softmax}(C)[o]$; write $q_\pi(o) := q(o \mid \pi)$ for this scored-policy outcome predictive, used in the remaining terms.
- *Ambiguity* is the expected likelihood entropy $\mathbb{E}_{q(s)}[H[p(o \mid s)]]$, the outcome uncertainty given the state.
- *Pragmatic value* is the expected log-preference $\mathbb{E}_{q_\pi(o)}[\ln p_C(o)]$ — the utility, exploitation term.
- *Epistemic value* is the state-outcome mutual information $H[q_\pi(o)] - \mathbb{E}_{q(s)}[H[p(o \mid s)]]$ — the expected information gain that drives exploration.

Expected free energy as the action-selection objective I

Because there is no sampling, the identity of eq. 19 holds to floating-point tolerance; ISC-19 (`expected_free_energy`) pins the residual of the decomposition to zero and pins each term's semantics independently (deterministic likelihoods give zero ambiguity; uninformative likelihoods give zero epistemic value; preference-matched predictions lower risk). fig. 7 visualizes the additive cost view and the signed pragmatic/epistemic waterfall terminating at $G(\pi)$ — a deterministic identity (Proposition 7), not a fitted result.

Sentinels also revise model *structure*. Bayesian model reduction (BMR) scores whether a reduced model — for example one that prunes a redundant location column by shrinking its concentration toward zero — has more evidence than the full model, *without re-running inference* (Friston & Penny via the post-hoc model optimization lineage (Friston and Penny 2011); the same Beta-function identity is their equation 13 (Friston et al. (2024))). Because the likelihood is shared, the reduced posterior is available in closed form, $\text{reduced\_post} = \text{post} + \text{reduced\_prior} - \text{prior}$, and the change in (negative) variational free energy is a difference of log multivariate Beta functions,

Bayesian model reduction for structure emergence I

$$
\begin{aligned}
\Delta F \;&=\; \ln B(\text{prior}) + \ln B(\text{reduced\_post}) - \ln B(\text{post}) - \ln B(\text{reduced\_prior}), \\
\ln B(a) \;&=\; \textstyle\sum_k \ln \Gamma(a_k) - \ln \Gamma(\sum_k a_k),
\end{aligned} \tag{3}
$$

computed in `bayesian_model_reduction.reduce`. A positive ΔF in eq. 3 means the reduced model carries more evidence — the pruned structure was redundant and should be adopted; a negative ΔF means the reduction destroyed support the data require. When the reduced prior equals the prior the score is identically zero in exact algebra, a zero point the suite pins to machine precision (ISC-20).

The emergence study of sec. 5.22 uses this operation over $n = 4$ candidate states. It contrasts a redundant reduction ($\Delta F = 3.68$ nats; adopt) with a supported one ($\Delta F = -27.67$ nats; reject) in fig. 11. This fixed-candidate algebraic comparison is deterministic, so it has no resampled sample or bootstrap interval.